TECHNICAL DOSSIER

Hexabot

Evolution of the Hormiga project into a hexapod spider robot with a new mechanical architecture, electronics, and inverse-kinematics control.

DOSSIER DATA

PROJECT LEAD
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STATUS
Technical redesign

REPOSITORY
github.com/RoboTech-URJC/HexaBot.git

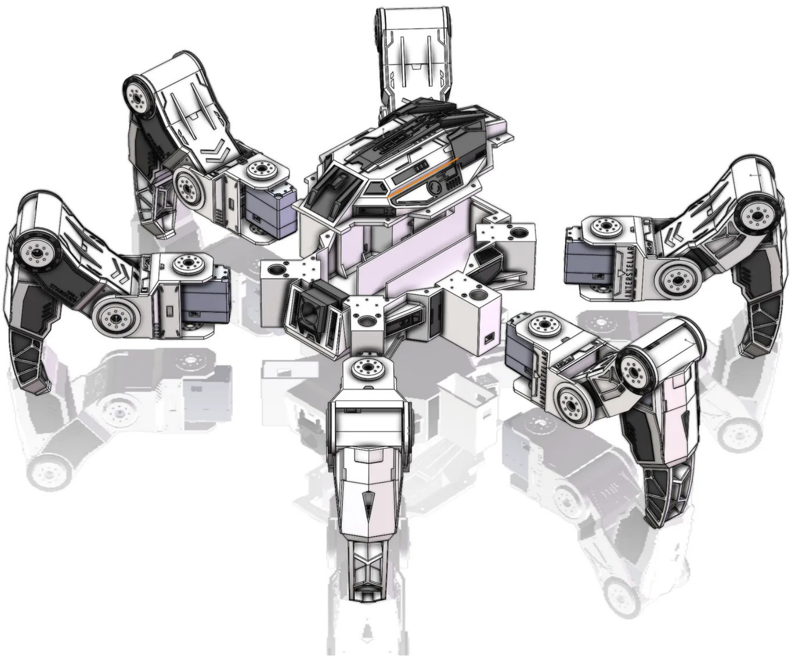
LAST UPDATE
June 2026

Project Description

The Hexabot project, also conceived as the new version of the association's robotic spider, builds on the already completed Hormiga project. The current goal is not to start from scratch, but to take advantage of that foundation and update it to obtain a hexapod platform that is easier to handle, more robust, and better prepared for future growth.

Hormiga project, the previous version used as the basis for the Hexabot redesign.

The main difference compared with the previous version is the change in mechanical architecture. The new design is based on a hexagonal hexapod frame, more suitable for handling and moving the robot than the frame used in Hormiga. In addition, the mechanical parts are being redesigned with higher quality, a better finish, and a structure better prepared for later modifications.



Hexapod frame used as a reference from Printables. Author: Tom Knox.

Objectives

General objective: redesign the robotic spider based on the Hormiga project, updating its mechanical architecture, electronics, and software to turn it into a more modular and expandable hexapod platform.

Specific objectives:

- Replace the previous frame with a hexagonal structure that is easier to handle and move.
- Redesign the mechanical parts to improve assembly quality and make future iterations easier.
- Update the electronic architecture on which the robot is based.
- Rewrite the control code so that movement is based on inverse kinematics.
- Prepare the frame for future expansions with computer vision and advanced processing hardware.

Work Completed During the Year

During this year, the work has focused on rethinking the robot architecture using the Hormiga project as a reference. Both the mechanical and electronic parts are being updated so that the new version is cleaner, easier to handle, and more suitable as a development platform.

On the mechanical side, a hexagonal chassis has been chosen. This decision aims to improve leg distribution and make both movement and robot handling easier during testing. The part design is also being reviewed so that the overall assembly reaches a higher quality than the previous version.

In parallel, the control software is being rewritten. The previous version worked in a more rigid way, based on pre-coded movements such as moving forward a specific number of steps. The new approach aims to make inverse kinematics easier to run, allowing movements to be defined in a more flexible and scalable way.

System Architecture

The planned architecture combines a hexapod mechanical base with renewed electronics and a more advanced control system. The robot will still depend on the coordination of its legs, but the goal is for movement not to be limited to fixed sequences and instead to be calculated more generally through inverse kinematics.

Inverse kinematics: the new code aims to replace control based on predefined steps with logic capable of calculating leg positions and movements in a more flexible way.

This change is important because it will make the robot a more useful platform for experimentation. Instead of programming each behavior in a closed way, the idea is to prepare a base that can receive new movement commands and adapt better to future developments.

Planned Development

In addition to the current redesign, one of the important objectives is to prepare the mechanical frame for future modifications. The idea is that, if Hexabot is later turned into a platform with perception capabilities, the design itself will already allow new components to be integrated without having to rebuild the entire robot.

Among these planned expansions is the possibility of adding computer vision. To do this, the robot could incorporate a Jetson Nano, cameras, or other sensors, so that in the future it can interpret its environment and not depend only on manual or preprogrammed movements.

Future Autonomous Capabilities

A particularly interesting line of evolution is turning Hexabot into an autonomous platform. With a processing unit such as a Jetson Nano and a camera system, the robot could eventually move by itself, make basic decisions based on visual information, and execute more complex behaviors.

This phase is not part of the first functional version, but it does shape the current design: the frame, electronics, and software must be planned so that they can grow in that direction.

Value of the Project for the Association

Hexabot has clear value both as a technical project and as a visual demonstration. As a robotic spider, it is eye-catching for outreach activities, but it also makes it possible to work with advanced robotics concepts: mechanical design, servomotor control, inverse kinematics, electronic integration, and, in later phases, computer vision.

In addition, since it starts from a previous project that has already been completed, it helps show how an association project can evolve, reuse previous lessons learned, and become a more powerful platform.

Next Goals

- Finish the design of the new hexagonal frame.
- Complete the redesign of the mechanical parts.
- Update the robot electronics.

- Rewrite and test the inverse-kinematics-based code.
- Validate the first movements of the new spider robot.
- Prepare the design for future expansions with Jetson Nano and cameras.